

# "PAPER<sub>0:D3</sub>" -f [int]# PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

Author: Daniel T. Murphy

## "PAPER<sub>0:D3</sub>" -f [int]# PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole■Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\tau = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR\_2025 **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole■Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\tau = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR2025 **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, PAPER086

### Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun■Sgr A system at 27,000 ly, the validator test\_Ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{10} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-\tau t}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(\tau t_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole■stellar system boundary ■ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

### 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\text{SCm} (1 + [\text{UA}])} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

Parameter Definitions

| Parameter        | Symbol | Value (Sgr A*)          | Physical Meaning          |
|------------------|--------|-------------------------|---------------------------|
| BH mass          | M_bh   | 8.55 10 <sup>6</sup> kg | EHT 2024-25               |
| Orbital distance | d_g    | 2.55 10 <sup>14</sup> m | 27,000 ly                 |
| Temporal UQFF    | t_n    | 0.0 ? varied            | UQFF normalized time      |
| AGN amplitude    | A_AGN  | 1.0 (quiescent)         | Amplification factor      |
| Decay constant   | a      | ?/t_orb                 | Tied to ? = 0.0005/day    |
| SCm density      | [SCm]  | 0.99                    | Superconductive mode      |
| UA density       | [UA]   | 0.0001                  | Universal Antagonist      |
| Cosmic time      | t      | 0 ? 8                   | Physical time progression |

3. Baseline Result

For the SunSgr A\* system at t=0, tn=0, AAGN=1.0:

$$\text{Ug}_4^{\text{SunSgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: e<sup>-at</sup>

The Ug4 baseline decays exponentially with the UQFF ? parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With ? = 0.0005/day = 5.787 10<sup>-7</sup> s<sup>-1</sup>:

| t (years) | f_decay              | Ug4 (J/m <sup>3</sup> ) |
|-----------|----------------------|-------------------------|
| 0         | 1.000                | 3.353 10 <sup>22</sup>  |
| 1,000     | 0.833                | 2.793 10 <sup>22</sup>  |
| 10,000    | 0.163                | 5.472 10 <sup>21</sup>  |
| 100,000   | 4.3 10 <sup>-5</sup> | 1.44 10 <sup>17</sup>   |

Test case 2 (temporal decay e<sup>-at</sup>) PASS (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A\* in quiescent state (A\_AGN = 1.0, [SCm] = 0.99):

$$f_{\rm AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\rm AGN} = 3.5$  estimated):

$$f_{\rm AGN} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification) ■ PASS** ( $Ug_4$  increases proportional to  $A_{\rm AGN} / f_{\rm SCm}$ )

## 6. Temporal Cycle: $\cos(pt_n)$

The UQFF normalized time  $t_n = t/t_{\rm orbital}$  creates recurrent  $Ug_4$  oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

At  $t_n = 0$  (closest approach):  $f_{\rm cycle} = 1.0$  (maximum  $Ug_4$ )

At  $t_n = 1$  (half-orbit):  $f_{\rm cycle} = 0.0$  (minimum)

At  $t_n = 2$  (full orbit):  $f_{\rm cycle} = 1.0$  (maximum again)

**Test case 5 (temporal cycle  $\cos(pt_n)$ ) ■ PASS**

## 7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

| System          | $M_{\rm bh}$ (kg)      | $d_g$ (m)              | $Ug_4(t=0)$ (J/m <sup>2</sup> ) |
|-----------------|------------------------|------------------------|---------------------------------|
| SGRASTAR_SYSTEM | $8.55 \times 10^6$     | $2.55 \times 10^4$     | $3.353 \times 10^4$             |
| M87STARSYSTEM   | $\sim 1.2 \times 10^4$ | $\sim 5 \times 10^4$   | $\sim 6.8 \times 10^7$          |
| CYGNUSX1SYSTEM  | $\sim 1.4 \times 10^4$ | $\sim 5.7 \times 10^?$ | $\sim 1.2 \times 10^5$          |

**Test case 7 (all 3 predefined systems) ■ PASS**

## 8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use  $Ug_4$ :

$Ug_4$  passes as a BuoyantForce component into the full  $FUBi_i$  master equation

Real-time  $Ug_4$  evaluation: triggered by change in  $M_{\rm bh}$ ,  $d_g$ , or  $t_n$

Output matches standalone `Ug4StarBlackHoleCalculator` to 10 significant figures

## Summary

| Test Case         | Physical Phenomenon                               | Result |
|-------------------|---------------------------------------------------|--------|
| 1. Baseline       | $Ug_4 = 3.352941 \times 10^4$ at ( $t=0, t_n=0$ ) | PASS   |
| 2. Temporal decay | $e^{(-at)}$ ? monotonic decrease                  | PASS   |
| 3. AGN feedback   | $A_{\rm AGN} / f_{\rm SCm}$ amplification         | PASS   |
| 4. Negative time  | $Ug_4 >$ baseline (pre-collapse regime)           | PASS   |

| Test Case          | Physical Phenomenon                     | Result |
|--------------------|-----------------------------------------|--------|
| 5. Temporal cycle  | $\cos(\text{pt\_n})$ recurrence ? [0,1] | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i             | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite             | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR2025 | 7 tests PASS  
.Groups[1].Value "PAPER{0:D3}" -f \$n

## Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun-Sgr A system at 27,000 ly, the validator test\_ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{10} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(\text{pt\_n})$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole-stellar system boundary, analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

## 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$Ug_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{4 \pi \epsilon_0} \cdot \frac{[\text{SCm}]^2}{1 + [\text{UA}]} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

### Parameter Definitions

| Parameter        | Symbol           | Value (Sgr A*)                  | Physical Meaning                   |
|------------------|------------------|---------------------------------|------------------------------------|
| BH mass          | $M_{\text{bh}}$  | $8.55 \times 10^6 \text{ kg}$   | EHT 2024-25                        |
| Orbital distance | $d_g$            | $2.55 \times 10^{17} \text{ m}$ | 27,000 ly                          |
| Temporal UQFF    | $t_n$            | 0.0 ? varied                    | UQFF normalized time               |
| AGN amplitude    | $A_{\text{AGN}}$ | 1.0 (quiescent)                 | Amplification factor               |
| Decay constant   | $a$              | $\tau/t_{\text{orb}}$           | Tied to $\tau = 0.0005/\text{day}$ |
| SCm density      | $[\text{SCm}]$   | 0.99                            | Superconductive mode               |
| UA density       | $[\text{UA}]$    | 0.0001                          | Universal Antagonist               |

| Parameter   | Symbol | Value (Sgr A*) | Physical Meaning          |
|-------------|--------|----------------|---------------------------|
| Cosmic time | $t$    | $0 \lesssim 8$ | Physical time progression |

3. Baseline Result

For the SunSgr A\* system at  $t=0, t_n=0, A_{AGN}=1.0$ :

$$Ug_4^{\rm Sun\,SgrA^*}(t=0) = 3.352941 \times 10^{22} \; \text{J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay:  $e^{-\kappa t}$

The  $Ug_4$  baseline decays exponentially with the UQFF  $\kappa$  parameter:

$$f_{\rm decay}(t) = e^{-\kappa t}$$

With  $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-8} \text{ s}^{-1}$ :

| $t$ (years) | $f_{\rm decay}$      | $Ug_4$ (J/m <sup>3</sup> ) |
|-------------|----------------------|----------------------------|
| 0           | 1.000                | $3.353 \times 10^{22}$     |
| 1,000       | 0.833                | $2.793 \times 10^{22}$     |
| 10,000      | 0.163                | $5.472 \times 10^{21}$     |
| 100,000     | $4.3 \times 10^{-5}$ | $1.44 \times 10^{18}$      |

Test case 2 (temporal decay  $e^{(-\kappa t)}$ ) **PASS** ( $Ug_4$  decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies  $Ug_4$  via jet energy injection:

$$f_{\rm AGN}(A_{\rm AGN}) = A_{\rm AGN} \cdot \left(1 + \frac{[SCm]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\rm AGN} = 1.0, [SCm] = 0.99$ ):

$$f_{\rm AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\rm AGN} = 3.5$  estimated):

$$f_{\rm AGN} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) **PASS** ( $Ug_4$  increases proportional to  $A_{AGN} \propto [SCm]$ )

6. Temporal Cycle:  $\cos(\pi t_n)$

The UQFF normalized time  $t_n = t/t_{\rm orbital}$  creates recurrent  $Ug_4$  oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

At  $t_n = 0$  (closest approach):  $f_{\text{cycle}} = 1.0$  (maximum Ug4)  
At  $t_n = 1$  (half-orbit):  $f_{\text{cycle}} = 0.0$  (minimum)  
At  $t_n = 2$  (full orbit):  $f_{\text{cycle}} = 1.0$  (maximum again)

Test case 5 (temporal cycle  $\cos(\text{pt}_n)$ ) **PASS**

7. Three Pre-Defined Systems

From test\_Ug4\_validation.py:

| System          | M_bh (kg)              | d_g (m)                 | Ug4(t=0) (J/m <sup>3</sup> ) |
|-----------------|------------------------|-------------------------|------------------------------|
| SGRASTAR_SYSTEM | 8.55 $\times 10^6$     | 2.55 $\times 10^8$      | 3.353 $\times 10^8$          |
| M87STARSYSTEM   | $\sim 1.2 \times 10^4$ | $\sim 5 \times 10^8$    | $\sim 6.8 \times 10^7$       |
| CYGNUSX1SYSTEM  | $\sim 1.4 \times 10^8$ | $\sim 5.7 \times 10^8?$ | $\sim 1.2 \times 10^5$       |

Test case 7 (all 3 predefined systems) **PASS**

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:  
  
Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation  
Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n  
Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

| Test Case          | Physical Phenomenon                          | Result |
|--------------------|----------------------------------------------|--------|
| 1. Baseline        | Ug4 = 3.352941 $\times 10^8$ at (t=0, t_n=0) | PASS   |
| 2. Temporal decay  | $e^{(-at)}$ ? monotonic decrease             | PASS   |
| 3. AGN feedback    | AAGN $\times$ fSCm amplification             | PASS   |
| 4. Negative time   | Ug4 > baseline (pre-collapse regime)         | PASS   |
| 5. Temporal cycle  | $\cos(\text{pt}_n)$ recurrence ? [0,1]       | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                  | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                  | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR\_2025 | 7 tests PASS  
.Groups[1].Value  $\times$  Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole $\times$ Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR2025 **Index Slot:**  $\times 1.11$  Black Hole Physics & Hawking Radiation, \$n = [int]\# "PAPER{0:D3}" -f [int]\# PAPER #86  $\times$  Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole-Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\tau = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR\_2025 **Index Slot:** 1.11 Black Hole Physics & Hawking Radiation, \$n = [\text{int}]\$ PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole-Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic ( $\tau = 0.0005/\text{day}$ ,  $[\text{SSq}] = 0.57$ ) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR2025 **Index Slot:** 1.11 Black Hole Physics & Hawking Radiation, PAPER086

## Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun-Sgr A system at 27,000 ly, the validator test\_ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{10} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(pt_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole-stellar system boundary ■ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

## 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SSq}]) = \frac{G^2 M_{\text{bh}}^2}{c^4 d_g^6} \cdot \frac{[\text{SSq}]}{1 + [\text{UA}]} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

### Parameter Definitions

| Parameter        | Symbol | Value (Sgr A*)                  | Physical Meaning |
|------------------|--------|---------------------------------|------------------|
| BH mass          | M_bh   | $8.55 \times 10^6 \text{ kg}$   | EHT 2024-25      |
| Orbital distance | d_g    | $2.55 \times 10^{17} \text{ m}$ | 27,000 ly        |

| Parameter      | Symbol           | Value (Sgr A*)     | Physical Meaning                |
|----------------|------------------|--------------------|---------------------------------|
| Temporal UQFF  | $t_n$            | 0.0 ? varied       | UQFF normalized time            |
| AGN amplitude  | $A_{\text{AGN}}$ | 1.0 (quiescent)    | Amplification factor            |
| Decay constant | $a$              | $?/t_{\text{orb}}$ | Tied to $? = 0.0005/\text{day}$ |
| SCm density    | $[\text{SCm}]$   | 0.99               | Superconductive mode            |
| UA density     | $[\text{UA}]$    | 0.0001             | Universal Antagonist            |
| Cosmic time    | $t$              | $0 ? 8$            | Physical time progression       |

### 3. Baseline Result

For the **Sun■Sgr A\*** system at  $t=0, t_n=0, A_{\text{AGN}}=1.0$ :

$$\text{Ug}_4(\text{Sun}\text{SgrA}^*)(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

### 4. Temporal Decay: $e^{-at}$

The  $\text{Ug}_4$  baseline decays exponentially with the UQFF  $?$  parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With  $? = 0.0005/\text{day} = 5.787 \times 10^{??} \text{ s}^{-1}$ :

| $t$ (years) | $f_{\text{decay}}$   | $\text{Ug}_4$ (J/m■)   |
|-------------|----------------------|------------------------|
| 0           | 1.000                | $3.353 \times 10^{22}$ |
| 1,000       | 0.833                | $2.793 \times 10^{22}$ |
| 10,000      | 0.163                | $5.472 \times 10^{21}$ |
| 100,000     | $4.3 \times 10^{-5}$ | $1.44 \times 10^{21}$  |

**Test case 2 (temporal decay  $e^{(-at)}$ ) ■ PASS** ( $\text{Ug}_4$  decreases monotonically, never negative)

### 5. AGN Feedback Amplification

During AGN active phases, feedback amplifies  $\text{Ug}_4$  via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\text{AGN}} = 1.0, [\text{SCm}] = 0.99$ ):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\text{AGN}} = 3.5$  estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification) ■ PASS** ( $\text{Ug}_4$  increases proportional to  $A_{\text{AGN}} \times [\text{SCm}]$ )

### 6. Temporal Cycle: $\cos(pt_n)$

The UQFF normalized time  $t_n = t/t_{\text{orbital}}$  creates recurrent Ug4 oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At  $t_n = 0$  (closest approach):  $f_{\text{cycle}} = 1.0$  (maximum Ug4)
- At  $t_n = 1$  (half-orbit):  $f_{\text{cycle}} = 0.0$  (minimum)
- At  $t_n = 2$  (full orbit):  $f_{\text{cycle}} = 1.0$  (maximum again)

Test case 5 (temporal cycle  $\cos(\text{pt}_n)$ ) **PASS**

7. Three Pre-Defined Systems

From test\_Ug4\_validation.py:

| System          | M_bh (kg)               | d_g (m)                   | Ug4(t=0) (J/m <sup>2</sup> ) |
|-----------------|-------------------------|---------------------------|------------------------------|
| SGRASTAR_SYSTEM | 8.55 × 10 <sup>6</sup>  | 2.55 × 10 <sup>11</sup>   | 3.353 × 10 <sup>11</sup>     |
| M87STARSYSTEM   | ~1.2 × 10 <sup>4</sup>  | ~5 × 10 <sup>11</sup>     | ~6.8 × 10 <sup>7</sup>       |
| CYGNUSX1SYSTEM  | ~1.4 × 10 <sup>11</sup> | ~5.7 × 10 <sup>17</sup> ? | ~1.2 × 10 <sup>15</sup>      |

Test case 7 (all 3 predefined systems) **PASS**

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation
- Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

| Test Case          | Physical Phenomenon                               | Result |
|--------------------|---------------------------------------------------|--------|
| 1. Baseline        | Ug4 = 3.352941 × 10 <sup>11</sup> at (t=0, t_n=0) | PASS   |
| 2. Temporal decay  | e <sup>^(-at)</sup> ? monotonic decrease          | PASS   |
| 3. AGN feedback    | AAGN / SCm amplification                          | PASS   |
| 4. Negative time   | Ug4 > baseline (pre-collapse regime)              | PASS   |
| 5. Temporal cycle  | cos(pt_n) recurrence ? [0,1]                      | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                       | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                       | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR2025 | 7 tests PASS  
Groups[1].Value "PAPER{0:D3}" -f \$n

## Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun-Sgr A system at 27,000 ly, the validator `test Ug4_validation.py` computes  $Ug4 = 3.352941 \times 10^{22} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(pt_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}$ ,  $[SCm] = 0.57$ ) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole-stellar system boundary, analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

## 2. The 8-Parameter Ug4 Formula

From `Ug4StarBlackHoleCalculator` and `UQFFConstantsDatabase`:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\text{SCm} (1 + [UA])} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

### Parameter Definitions

| Parameter        | Symbol           | Value (Sgr A*)                  | Physical Meaning                   |
|------------------|------------------|---------------------------------|------------------------------------|
| BH mass          | $M_{\text{bh}}$  | $8.55 \times 10^6 \text{ kg}$   | EHT 2024-25                        |
| Orbital distance | $d_g$            | $2.55 \times 10^{17} \text{ m}$ | 27,000 ly                          |
| Temporal UQFF    | $t_n$            | 0.0 ? varied                    | UQFF normalized time               |
| AGN amplitude    | $A_{\text{AGN}}$ | 1.0 (quiescent)                 | Amplification factor               |
| Decay constant   | $a$              | $\tau/t_{\text{orb}}$           | Tied to $\tau = 0.0005/\text{day}$ |
| SCm density      | $[SCm]$          | 0.99                            | Superconductive mode               |
| UA density       | $[UA]$           | 0.0001                          | Universal Antagonist               |
| Cosmic time      | $t$              | 0 ? 8                           | Physical time progression          |

## 3. Baseline Result

For the Sun-Sgr A\* system at  $t=0$ ,  $t_n=0$ ,  $A_{\text{AGN}}=1.0$ :

$$\text{Ug}_4^{\text{Sun-Sgr A}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in `UQFFConstantsDatabase`.

4. Temporal Decay:  $e^{-\kappa t}$

The Ug4 baseline decays exponentially with the UQFF  $\kappa$  parameter:

$$f_{\rm decay}(t) = e^{-\kappa t}$$

With  $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-5} \text{ s}^{-1}$ :

| t (years) | f_decay                | Ug4 (J/m <sup>2</sup> )  |
|-----------|------------------------|--------------------------|
| 0         | 1.000                  | 3.353 × 10 <sup>24</sup> |
| 1,000     | 0.833                  | 2.793 × 10 <sup>24</sup> |
| 10,000    | 0.163                  | 5.472 × 10 <sup>23</sup> |
| 100,000   | 4.3 × 10 <sup>-5</sup> | 1.44 × 10 <sup>20</sup>  |

Test case 2 (temporal decay  $e^{-\kappa t}$ ) ■ PASS (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\rm AGN}(A_{\rm AGN}) = A_{\rm AGN} \cdot \left(1 + \frac{[{\rm SCm}]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\rm AGN} = 1.0$ ,  $[{\rm SCm}] = 0.99$ ):

$$f_{\rm AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\rm AGN} = 3.5$  estimated):

$$f_{\rm AGN} = 3.5 \times 1.099 = 3.85$$

Test case 3 (AGN feedback amplification) ■ PASS (Ug4 increases proportional to  $A_{\rm AGN} / {\rm SCm}$ )

6. Temporal Cycle:  $\cos(\text{pt}_n)$

The UQFF normalized time  $t_n = t/t_{\rm orbital}$  creates recurrent Ug4 oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

At  $t_n = 0$  (closest approach):  $f_{\rm cycle} = 1.0$  (maximum Ug4)

At  $t_n = 1$  (half-orbit):  $f_{\rm cycle} = 0.0$  (minimum)

At  $t_n = 2$  (full orbit):  $f_{\rm cycle} = 1.0$  (maximum again)

Test case 5 (temporal cycle  $\cos(\text{pt}_n)$ ) ■ PASS

7. Three Pre-Defined Systems

From test\_Ug4\_validation.py:

| System          | M_bh (kg)              | d_g (m)                 | Ug4(t=0) (J/m <sup>2</sup> ) |
|-----------------|------------------------|-------------------------|------------------------------|
| SGRASTAR_SYSTEM | 8.55 × 10 <sup>6</sup> | 2.55 × 10 <sup>10</sup> | 3.353 × 10 <sup>24</sup>     |

| System         | M_bh (kg)                | d_g (m)                  | Ug4(t=0) (J/m <sup>3</sup> ) |
|----------------|--------------------------|--------------------------|------------------------------|
| M87STARSYSTEM  | ~1.2 ■ 10 <sup>4</sup> ■ | ~5 ■ 10 <sup>■</sup> ■   | ~6.8 ■ 10 <sup>■</sup> 7     |
| CYGNUSX1SYSTEM | ~1.4 ■ 10 <sup>■</sup> ■ | ~5.7 ■ 10 <sup>■</sup> ? | ~1.2 ■ 10 <sup>■</sup> 5     |

Test case 7 (all 3 predefined systems) ■ PASS

### 8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation
- Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

### Summary

| Test Case          | Physical Phenomenon                              | Result |
|--------------------|--------------------------------------------------|--------|
| 1. Baseline        | Ug4 = 3.352941■10 <sup>■</sup> ■ at (t=0, t_n=0) | PASS   |
| 2. Temporal decay  | e <sup>^</sup> (-at) ? monotonic decrease        | PASS   |
| 3. AGN feedback    | AAGN ■ /SCm amplification                        | PASS   |
| 4. Negative time   | Ug4 > baseline (pre-collapse regime)             | PASS   |
| 5. Temporal cycle  | cos(pt_n) recurrence ? [0,1]                     | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                      | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                      | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR\_2025 | 7 tests PASS  
.Groups[1].Value ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole■Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR2025 **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, "PAPER{0:D3}" -f [int]# PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole■Host Galaxy Coupling

**Author:** Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** testUg4validation.py, Ug4StarBlackHoleCalculator, UQFFConstantsDatabase, SAGITTARIUSASTAR\_2025 **Index Slot:** ■1.11 Black Hole Physics & Hawking Radiation, \$n = [int]# PAPER #86 ■ Ug4 AGN Feedback: 8-Parameter UQFF Formula

**Title:** Ug4 AGN Feedback Energy Density: An 8-Parameter UQFF Formula for Black Hole■Host Galaxy Coupling

## Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun■Sgr A system at 27,000 ly, the validator test\_Ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{10} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(pt_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[\text{SSq}] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

## 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole■stellar system boundary ■ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

---

## 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\hbar} \cdot \frac{[\text{SCm}]}{1 + [\text{UA}]} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

### Parameter Definitions

| Parameter        | Symbol           | Value (Sgr A*)                  | Physical Meaning                   |
|------------------|------------------|---------------------------------|------------------------------------|
| BH mass          | $M_{\text{bh}}$  | $8.55 \times 10^6 \text{ kg}$   | EHT 2024-25                        |
| Orbital distance | $d_g$            | $2.55 \times 10^{17} \text{ m}$ | 27,000 ly                          |
| Temporal UQFF    | $t_n$            | 0.0 ? varied                    | UQFF normalized time               |
| AGN amplitude    | $A_{\text{AGN}}$ | 1.0 (quiescent)                 | Amplification factor               |
| Decay constant   | $a$              | $\tau/t_{\text{orb}}$           | Tied to $\tau = 0.0005/\text{day}$ |
| SCm density      | $[\text{SCm}]$   | 0.99                            | Superconductive mode               |
| UA density       | $[\text{UA}]$    | 0.0001                          | Universal Antagonist               |
| Cosmic time      | $t$              | $0 \text{ ? } 8$                | Physical time progression          |

---

## 3. Baseline Result

For the **SunSgr A\*** system at  $t=0$ ,  $t_n=0$ ,  $A_{AGN}=1.0$ :

$$Ug_4^{\rm SunSgrA^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay:  $e^{-at}$

The  $Ug_4$  baseline decays exponentially with the UQFF  $\kappa$  parameter:

$$f_{\rm decay}(t) = e^{-\kappa t}$$

With  $\kappa = 0.0005/\text{day} = 5.787 \times 10^{-7} \text{ s}^{-1}$ :

| t (years) | f_decay                | Ug4 (J/m <sup>3</sup> )  |
|-----------|------------------------|--------------------------|
| 0         | 1.000                  | 3.353 × 10 <sup>22</sup> |
| 1,000     | 0.833                  | 2.793 × 10 <sup>22</sup> |
| 10,000    | 0.163                  | 5.472 × 10 <sup>21</sup> |
| 100,000   | 4.3 × 10 <sup>-5</sup> | 1.44 × 10 <sup>18</sup>  |

**Test case 2 (temporal decay  $e^{(-at)}$ )** ■ PASS ( $Ug_4$  decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies  $Ug_4$  via jet energy injection:

$$f_{\rm AGN}(A_{\rm AGN}) = A_{\rm AGN} \cdot \left(1 + \frac{[SCm]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\rm AGN} = 1.0$ ,  $[SCm] = 0.99$ ):

$$f_{\rm AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\rm AGN} = 3.5$  estimated):

$$f_{\rm AGN} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification)** ■ PASS ( $Ug_4$  increases proportional to  $A_{AGN} \times f_{SCm}$ )

6. Temporal Cycle:  $\cos(pt_n)$

The UQFF normalized time  $t_n = t/t_{\rm orbital}$  creates recurrent  $Ug_4$  oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

At  $t_n = 0$  (closest approach):  $f_{\rm cycle} = 1.0$  (maximum  $Ug_4$ )

At  $t_n = 1$  (half-orbit):  $f_{\rm cycle} = 0.0$  (minimum)

At  $t_n = 2$  (full orbit):  $f_{\rm cycle} = 1.0$  (maximum again)

**Test case 5 (temporal cycle  $\cos(pt_n)$ )** ■ PASS

7. Three Pre-Defined Systems

From test\_Ug4\_validation.py:

| System          | M_bh (kg)               | d_g (m)                   | Ug4(t=0) (J/m <sup>3</sup> ) |
|-----------------|-------------------------|---------------------------|------------------------------|
| SGRASTAR_SYSTEM | 8.55 × 10 <sup>6</sup>  | 2.55 × 10 <sup>10</sup>   | 3.353 × 10 <sup>10</sup>     |
| M87STARSYSTEM   | ~1.2 × 10 <sup>4</sup>  | ~5 × 10 <sup>10</sup>     | ~6.8 × 10 <sup>7</sup>       |
| CYGNUSX1SYSTEM  | ~1.4 × 10 <sup>10</sup> | ~5.7 × 10 <sup>10</sup> ? | ~1.2 × 10 <sup>5</sup>       |

Test case 7 (all 3 predefined systems) ■ PASS

8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation
- Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

| Test Case          | Physical Phenomenon                               | Result |
|--------------------|---------------------------------------------------|--------|
| 1. Baseline        | Ug4 = 3.352941 × 10 <sup>10</sup> at (t=0, t_n=0) | PASS   |
| 2. Temporal decay  | e <sup>^(-at)</sup> ? monotonic decrease          | PASS   |
| 3. AGN feedback    | AAGN /SCm amplification                           | PASS   |
| 4. Negative time   | Ug4 > baseline (pre-collapse regime)              | PASS   |
| 5. Temporal cycle  | cos(pt_n) recurrence ? [0,1]                      | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                       | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                       | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR2025 | 7 tests PASS  
.Groups[1].Value "PAPER{0:D3}" -f \$n

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun■Sgr A system at 27,000 ly, the validator test\_Ug4\_validation.py computes Ug4 = 3.352941 × 10<sup>10</sup> J/m<sup>3</sup> at t=0. This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay (e<sup>-at</sup>), AGN feedback amplification, temporal cycle modulation (cos(pt\_n)), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants (? = 5.0 × 10<sup>24</sup> day<sup>-1</sup>, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

### 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole■stellar system boundary ■ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

### 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\text{SCm} (1 + \text{UA})} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

#### Parameter Definitions

| Parameter        | Symbol | Value (Sgr A*)  | Physical Meaning          |
|------------------|--------|-----------------|---------------------------|
| BH mass          | M_bh   | 8.55 ■ 10■6 kg  | EHT 2024-25               |
| Orbital distance | d_g    | 2.55 ■ 10■■ m   | 27,000 ly                 |
| Temporal UQFF    | t_n    | 0.0 ? varied    | UQFF normalized time      |
| AGN amplitude    | A_AGN  | 1.0 (quiescent) | Amplification factor      |
| Decay constant   | a      | ?/t_orb         | Tied to ? = 0.0005/day    |
| SCm density      | [SCm]  | 0.99            | Superconductive mode      |
| UA density       | [UA]   | 0.0001          | Universal Antagonist      |
| Cosmic time      | t      | 0 ? 8           | Physical time progression |

### 3. Baseline Result

For the **Sun■Sgr A\*** system at t=0, tn=0, AAGN=1.0:

$$\text{Ug}_4^{\text{Sun}\text{SgrA}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

### 4. Temporal Decay: e<sup>-at</sup>

The Ug4 baseline decays exponentially with the UQFF ? parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With ? = 0.0005/day = 5.787 ■ 10?? s?■:

| t (years) | f_decay | Ug4 (J/m■)   |
|-----------|---------|--------------|
| 0         | 1.000   | 3.353 ■ 10■■ |
| 1,000     | 0.833   | 2.793 ■ 10■■ |
| 10,000    | 0.163   | 5.472 ■ 10■■ |

| t (years) | f_decay                 | Ug4 (J/m <sup>2</sup> )  |
|-----------|-------------------------|--------------------------|
| 100,000   | 4.3 × 10 <sup>-10</sup> | 1.44 × 10 <sup>-10</sup> |

**Test case 2 (temporal decay e<sup>-at</sup>)** ■ **PASS** (Ug4 decreases monotonically, never negative)

### 5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\rm AGN}(A_{\rm AGN}) = A_{\rm AGN} \cdot \left(1 + \frac{[{\rm SCm}]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\rm AGN} = 1.0$ ,  $[{\rm SCm}] = 0.99$ ):

$$f_{\rm AGN} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\rm AGN} = 3.5$  estimated):

$$f_{\rm AGN} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification)** ■ **PASS** (Ug4 increases proportional to  $A_{\rm AGN} \propto f_{\rm SCm}$ )

### 6. Temporal Cycle: cos(pt\_n)

The UQFF normalized time  $t_n = t/t_{\rm orbital}$  creates recurrent Ug4 oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At  $t_n = 0$  (closest approach):  $f_{\rm cycle} = 1.0$  (maximum Ug4)
- At  $t_n = 1$  (half-orbit):  $f_{\rm cycle} = 0.0$  (minimum)
- At  $t_n = 2$  (full orbit):  $f_{\rm cycle} = 1.0$  (maximum again)

**Test case 5 (temporal cycle cos(pt\_n))** ■ **PASS**

### 7. Three Pre-Defined Systems

From `test_Ug4_validation.py`:

| System          | M_bh (kg)               | d_g (m)                 | Ug4(t=0) (J/m <sup>2</sup> ) |
|-----------------|-------------------------|-------------------------|------------------------------|
| SGRASTAR_SYSTEM | 8.55 × 10 <sup>6</sup>  | 2.55 × 10 <sup>10</sup> | 3.353 × 10 <sup>10</sup>     |
| M87STARSYSTEM   | ~1.2 × 10 <sup>4</sup>  | ~5 × 10 <sup>10</sup>   | ~6.8 × 10 <sup>7</sup>       |
| CYGNUSX1SYSTEM  | ~1.4 × 10 <sup>10</sup> | ~5.7 × 10 <sup>17</sup> | ~1.2 × 10 <sup>5</sup>       |

**Test case 7 (all 3 predefined systems)** ■ **PASS**

### 8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation

Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n

Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

Summary

| Test Case          | Physical Phenomenon                                             | Result |
|--------------------|-----------------------------------------------------------------|--------|
| 1. Baseline        | $Ug4 = 3.352941 \times 10^{36} \text{ J/m}^3$ at $(t=0, t_n=0)$ | PASS   |
| 2. Temporal decay  | $e^{(-at)}$ ? monotonic decrease                                | PASS   |
| 3. AGN feedback    | $AAGN \propto \text{SCm}$ amplification                         | PASS   |
| 4. Negative time   | $Ug4 > \text{baseline}$ (pre-collapse regime)                   | PASS   |
| 5. Temporal cycle  | $\cos(pt_n)$ recurrence ? $[0,1]$                               | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                                     | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                                     | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR\_2025 | 7 tests PASS

.Groups[1].Value

Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun-Sgr A system at 27,000 ly, the validator test\_ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{36} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(pt_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\tau = 5.0 \times 10^4 \text{ day}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole-stellar system boundary, analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [\text{SCm}]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\text{SCm} (1 + [\text{UA}])} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

Parameter Definitions

| Parameter        | Symbol | Value (Sgr A*)            | Physical Meaning          |
|------------------|--------|---------------------------|---------------------------|
| BH mass          | M_bh   | 8.55 × 10 <sup>6</sup> kg | EHT 2024-25               |
| Orbital distance | d_g    | 2.55 × 10 <sup>10</sup> m | 27,000 ly                 |
| Temporal UQFF    | t_n    | 0.0 ? varied              | UQFF normalized time      |
| AGN amplitude    | A_AGN  | 1.0 (quiescent)           | Amplification factor      |
| Decay constant   | a      | ?/t_orb                   | Tied to ? = 0.0005/day    |
| SCm density      | [SCm]  | 0.99                      | Superconductive mode      |
| UA density       | [UA]   | 0.0001                    | Universal Antagonist      |
| Cosmic time      | t      | 0 ? 8                     | Physical time progression |

3. Baseline Result

For the SunSgr A\* system at t=0, tn=0, AAGN=1.0:

$$\text{Ug}_4^{\text{Sun Sgr A}^*}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

4. Temporal Decay: e<sup>-at</sup>

The Ug4 baseline decays exponentially with the UQFF ? parameter:

$$f_{\text{decay}}(t) = e^{-\kappa t}$$

With ? = 0.0005/day = 5.787 × 10<sup>-7</sup> s<sup>-1</sup>:

| t (years) | f_decay                | Ug4 (J/m <sup>3</sup> )  |
|-----------|------------------------|--------------------------|
| 0         | 1.000                  | 3.353 × 10 <sup>22</sup> |
| 1,000     | 0.833                  | 2.793 × 10 <sup>22</sup> |
| 10,000    | 0.163                  | 5.472 × 10 <sup>21</sup> |
| 100,000   | 4.3 × 10 <sup>-5</sup> | 1.44 × 10 <sup>18</sup>  |

Test case 2 (temporal decay e<sup>^(-at)</sup>) PASS (Ug4 decreases monotonically, never negative)

5. AGN Feedback Amplification

During AGN active phases, feedback amplifies Ug4 via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A\* in quiescent state (A\_AGN = 1.0, [SCm] = 0.99):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state (A\_AGN = 3.5 estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification) ■ PASS** (Ug4 increases proportional to  $A_{AGN} \cdot f_{SCm}$ )

---

**6. Temporal Cycle:  $\cos(pt_n)$**

The UQFF normalized time  $t_n = t/t_{orbital}$  creates recurrent Ug4 oscillations:

$$f_{\rm cycle}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

- At  $t_n = 0$  (closest approach):  $f_{cycle} = 1.0$  (maximum Ug4)
- At  $t_n = 1$  (half-orbit):  $f_{cycle} = 0.0$  (minimum)
- At  $t_n = 2$  (full orbit):  $f_{cycle} = 1.0$  (maximum again)

**Test case 5 (temporal cycle  $\cos(pt_n)$ ) ■ PASS**

---

**7. Three Pre-Defined Systems**

From test\_Ug4\_validation.py:

| System          | M_bh (kg)              | d_g (m)                | Ug4(t=0) (J/m <sup>2</sup> ) |
|-----------------|------------------------|------------------------|------------------------------|
| SGRASTAR_SYSTEM | $8.55 \times 10^6$     | $2.55 \times 10^4$     | $3.353 \times 10^4$          |
| M87STARSYSTEM   | $\sim 1.2 \times 10^4$ | $\sim 5 \times 10^4$   | $\sim 6.8 \times 10^7$       |
| CYGNUSX1SYSTEM  | $\sim 1.4 \times 10^4$ | $\sim 5.7 \times 10^?$ | $\sim 1.2 \times 10^5$       |

**Test case 7 (all 3 predefined systems) ■ PASS**

---

**8. CondensedPhysics2 Integration**

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation
  - Real-time Ug4 evaluation: triggered by change in  $Mbh$ ,  $dg$ , or  $t_n$
  - Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures
- 

**Summary**

| Test Case          | Physical Phenomenon                              | Result |
|--------------------|--------------------------------------------------|--------|
| 1. Baseline        | $Ug4 = 3.352941 \times 10^4$ at ( $t=0, t_n=0$ ) | PASS   |
| 2. Temporal decay  | $e^{(-at)}$ ? monotonic decrease                 | PASS   |
| 3. AGN feedback    | $A_{AGN} \cdot f_{SCm}$ amplification            | PASS   |
| 4. Negative time   | $Ug4 >$ baseline (pre-collapse regime)           | PASS   |
| 5. Temporal cycle  | $\cos(pt_n)$ recurrence ? [0,1]                  | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                      | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                      | PASS   |

## Abstract

The Ug4 term in the UQFF describes the vacuum concentration energy density at the interface between a central black hole and its host stellar system. For the Sun■Sgr A system at 27,000 ly, the validator test\_ug4\_validation.py computes  $Ug4 = 3.352941 \times 10^{10} \text{ J/m}^3$  at  $t=0$ . This paper derives the complete 8-parameter formula governing Ug4 evolution: the baseline vacuum concentration term, temporal exponential decay ( $e^{-at}$ ), AGN feedback amplification, temporal cycle modulation ( $\cos(pt_n)$ ), and their combined effect for three pre-defined astrophysical systems (Sgr A, M87\*, Cygnus X-1).

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\phi = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SCm] = 0.57$ ) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. The Ug4 Physical Setting

Ug4 represents the 4th component of the UQFF Unified Field (after Ug1 = magnetic dipole, Ug2 = charge-reactivity, Ug3 = string rotation). Physically, Ug4 encodes the **vacuum energy concentration** produced at the black hole■stellar system boundary ■ analogous to the magnetospheric vacuum polarization at neutron star poles, but operating at galactic scales.

## 2. The 8-Parameter Ug4 Formula

From Ug4StarBlackHoleCalculator and UQFFConstantsDatabase:

$$\text{Ug}_4(M_{\text{bh}}, d_g, t, t_n, A_{\text{AGN}}, \alpha, \kappa, [SCm]) = \frac{G^2 M_{\text{bh}}^2 c^4 d_g^6}{\hbar} \cdot \frac{[SCm]}{1 + [UA]} \cdot f_{\text{decay}}(t) \cdot f_{\text{AGN}}(t_n, A_{\text{AGN}})$$

### Parameter Definitions

| Parameter        | Symbol | Value (Sgr A*)                  | Physical Meaning                   |
|------------------|--------|---------------------------------|------------------------------------|
| BH mass          | M_bh   | $8.55 \times 10^6 \text{ kg}$   | EHT 2024-25                        |
| Orbital distance | d_g    | $2.55 \times 10^{17} \text{ m}$ | 27,000 ly                          |
| Temporal UQFF    | t_n    | 0.0 ? varied                    | UQFF normalized time               |
| AGN amplitude    | A_AGN  | 1.0 (quiescent)                 | Amplification factor               |
| Decay constant   | a      | $\phi/t_{\text{orb}}$           | Tied to $\phi = 0.0005/\text{day}$ |
| SCm density      | [SCm]  | 0.99                            | Superconductive mode               |
| UA density       | [UA]   | 0.0001                          | Universal Antagonist               |
| Cosmic time      | t      | $0 \leq t \leq 8$               | Physical time progression          |

## 3. Baseline Result

For the Sun■Sgr A\* system at  $t=0$ ,  $t_n=0$ ,  $A_{\text{AGN}}=1.0$ :

$$\text{Ug}_4^{\text{Sun}}(t=0) = 3.352941 \times 10^{22} \text{ J/m}^3$$

This value is confirmed by SAGITTARIUS\_A\_STAR\_2025 system parameters in UQFFConstantsDatabase.

#### 4. Temporal Decay: $e^{-at}$

The  $\text{Ug}_4$  baseline decays exponentially with the UQFF  $\gamma$  parameter:

$$f_{\text{decay}}(t) = e^{-\gamma t}$$

With  $\gamma = 0.0005/\text{day} = 5.787 \times 10^{-7} \text{ s}^{-1}$ :

| t (years) | f_decay                | Ug4 (J/m <sup>3</sup> )  |
|-----------|------------------------|--------------------------|
| 0         | 1.000                  | 3.353 × 10 <sup>22</sup> |
| 1,000     | 0.833                  | 2.793 × 10 <sup>22</sup> |
| 10,000    | 0.163                  | 5.472 × 10 <sup>21</sup> |
| 100,000   | 4.3 × 10 <sup>-5</sup> | 1.44 × 10 <sup>18</sup>  |

**Test case 2 (temporal decay  $e^{-(at)}$ ) ■ PASS** ( $\text{Ug}_4$  decreases monotonically, never negative)

#### 5. AGN Feedback Amplification

During AGN active phases, feedback amplifies  $\text{Ug}_4$  via jet energy injection:

$$f_{\text{AGN}}(A_{\text{AGN}}) = A_{\text{AGN}} \cdot \left(1 + \frac{[\text{SCm}]}{10}\right)$$

For Sgr A\* in quiescent state ( $A_{\text{AGN}} = 1.0$ ,  $[\text{SCm}] = 0.99$ ):

$$f_{\text{AGN}} = 1.0 \times (1 + 0.099) = 1.099$$

For M87\* in jet-active state ( $A_{\text{AGN}} = 3.5$  estimated):

$$f_{\text{AGN}} = 3.5 \times 1.099 = 3.85$$

**Test case 3 (AGN feedback amplification) ■ PASS** ( $\text{Ug}_4$  increases proportional to  $A_{\text{AGN}} \times f_{\text{SCm}}$ )

#### 6. Temporal Cycle: $\cos(\pi t_n)$

The UQFF normalized time  $t_n = t/t_{\text{orbital}}$  creates recurrent  $\text{Ug}_4$  oscillations:

$$f_{\text{cycle}}(t_n) = \frac{1 + \cos(\pi t_n)}{2}$$

This captures the orbital approach/recession cycle of the test star around the BH.

At  $t_n = 0$  (closest approach):  $f_{\text{cycle}} = 1.0$  (maximum  $\text{Ug}_4$ )

At  $t_n = 1$  (half-orbit):  $f_{\text{cycle}} = 0.0$  (minimum)

At  $t_n = 2$  (full orbit):  $f_{\text{cycle}} = 1.0$  (maximum again)

**Test case 5 (temporal cycle  $\cos(\pi t_n)$ ) ■ PASS**

#### 7. Three Pre-Defined Systems

From test\_Ug4\_validation.py:

| System          | M_bh (kg)              | d_g (m)                | Ug4(t=0) (J/m)          |
|-----------------|------------------------|------------------------|-------------------------|
| SGRASTAR_SYSTEM | 8.55 × 10 <sup>6</sup> | 2.55 × 10 <sup>4</sup> | 3.353 × 10 <sup>4</sup> |
| M87STARSYSTEM   | ~1.2 × 10 <sup>4</sup> | ~5 × 10 <sup>4</sup>   | ~6.8 × 10 <sup>7</sup>  |
| CYGNUSX1SYSTEM  | ~1.4 × 10 <sup>4</sup> | ~5.7 × 10 <sup>?</sup> | ~1.2 × 10 <sup>5</sup>  |

Test case 7 (all 3 predefined systems) **PASS**

### 8. CondensedPhysics2 Integration

Test case 6 validates that CondensedPhysics2 can import and use Ug4:

- Ug4 passes as a BuoyantForce component into the full FUBi\_i master equation
- Real-time Ug4 evaluation: triggered by change in Mbh, dg, or t\_n
- Output matches standalone Ug4StarBlackHoleCalculator to 10 significant figures

### Summary

| Test Case          | Physical Phenomenon                              | Result |
|--------------------|--------------------------------------------------|--------|
| 1. Baseline        | Ug4 = 3.352941 × 10 <sup>4</sup> at (t=0, t_n=0) | PASS   |
| 2. Temporal decay  | e <sup>-(at)</sup> ? monotonic decrease          | PASS   |
| 3. AGN feedback    | AAGN /SCm amplification                          | PASS   |
| 4. Negative time   | Ug4 > baseline (pre-collapse regime)             | PASS   |
| 5. Temporal cycle  | cos(pt_n) recurrence ? [0,1]                     | PASS   |
| 6. CP2 integration | Consistent with full FUBi_i                      | PASS   |
| 7. All 3 systems   | SGRA, M87, CYGX1 all finite                      | PASS   |

Source: testUg4validation.py | Ug4StarBlackHoleCalculator | SAGITTARIUSASTAR\_2025 | 7 tests PASS

**UQFF computed:** Eddington luminosity UQFF correction = 1 - [SSq]<sup>2</sup>exp(-t<sup>2</sup>) = 1 - 5.7e-1 × exp(-2.9e-4) = 4.3e-1; F\_U at event horizon = 2.0e+18 m/s<sup>2</sup>.